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Internship Report

Phage-Host Classification Using Conformal Prediction????

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Abstract

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Contents

Acknowledgements	3
1 Introduction	1
2 Background & State of the Art	3
2.1 Bacteriophages and Host-Prediction Problem	3
2.2 State of the Art	4
2.3 State of the Art in Computational Host Prediction	4
2.3.1 Sequence-Similarity and Homology-Based Heuristics	4
2.3.2 Protein-Content and Classical Machine Learning Paradigm	5
2.3.3 Deep Language Models and Protein Embeddings	5
2.4 The Critical Deficiency: Failure of Uncertainty Quantification	6
2.4.1 Why Bayesian Formulations Are Unsuitable	6
2.4.2 Why Standard Calibration Methods Are Insufficient	6
2.5 Conformal Prediction	7
2.5.1 Setup and nonconformity scores	7
2.5.2 prediction set and coverage guarantee	7
2.5.3 Split conformal prediction	7
2.6 The CSA Method	7
2.6.1 Motivation	7
2.6.2 Two-stage calibration	8
2.6.3 The envelope concept	8
2.6.4 Limitations of the original method	8
3 Methods	9
3.1 Problem Formalization	9
3.2 Non Convex envelope for classification	9
3.3 Score Transformation	9
3.4 Envelope Constriction	9
3.4.1 CSA Two-Stage Calibration	9
3.4.2 τ Computation	10
3.4.3 Radial Method	10
3.4.4 Strip Method	10
3.4.5 1D	10
3.4.6 Class-Specific $\hat{t}$ Calculation	10
3.5 $\hat{t}$ Calculation	10

4	Data	11
4.1	Data Description	11
4.2	Protein to Phage Aggregation	11
4.2.1	NaN handling after aggregation	11
4.3	Train-Test Split	12
5	Experiments and Results	13
5.1	Synthetic Data Validation	13
5.1.1	Purpose and setup	13
5.1.2	2D Experiments	13
5.1.3	Extension to K Dimensions	13
5.1.4	Multiple Hosts	13
5.2	Gram Type Classification	14
5.2.1	Experimental Setup	14
5.2.2	Results	14
5.3	Host Classification	14
6	Conclusion and Future Work	15
6.1	Summary.....	15
6.2	Limitations	15

1 Introduction

The rapid rise of antibiotic-resistant bacteria is one of the major challenges that global medicine is facing today. To overcome this issue, researchers are going to alternative solutions, one of which is phage therapy. Phage therapy relies on using the naturally occurring viruses to target and destroy the harmful bacteria.

However, To make this therapy a viable alternative, we first need to understand the basic biology of these viruses. Bacteriophages, or phages are viruses that infect bacterial hosts that and are one of the most abundant entities on the planet. Each phage contains a collection of structural and functional proteins that collectively determine the bacterial strains that the phage can infect - this property is called the host range of the phage.

These phages are highly host-specific: a phage which infects one bacterial host may have very little to no effect on another. This particular property of theirs has naturally generated interest in modelling and applications in the context of..... . Therefore a significant problem in the phage biology is the prediction of the bacterial hosts associated with a given phage. Accurate host-predictions can have applications in

The host specificity of phages is one of phage therapy's greatest advantage, but at the same time, it also brings a very significant practical challenge. While it allows us to target specific harmful bacteria without destroying the beneficial ones, it also causes problems in finding exactly the appropriate phage for a specific infection. Doing this by only experimental screening can be time-consuming and expensive. This motivates the development of reliable computational models that are capable of predicting these phage-host interactions directly from the protein level information.

The existing computational tools, however have a very important limitation: they typically only produce point predictions, which are often represented as a scalar confidence score between 0 and 1 with no associated measure of uncertainty. This lack of having an uncertainty quantification is problematic in clinical contexts, where failing to identify a true host could mean missing a life-saving treatment. Consequently, to build trust in automated host prediction, we need a mathematical framework that is capable of quantifying its own predictive uncertainty and provides reliable confidence guarantees.

The primary objective of this report is to address this limitation by developing an uncertainty-aware framework for phage-host prediction using **Conformal Prediction**. Conformal prediction provides a distribution-free framework that allows us to transform standard point predictions into prediction sets that are guaranteed to contain the true bacterial host with a user-defined confidence level.

In this project, we build our phage representations by aggregating collections of protein-level functional scores. This process naturally produced high-dimensional, structured score vectors, as different proteins and functions vary in their data availability. These score vectors also often contain missing or noisy information. To handle these chal-

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lenges, we explored geometric conformal prediction methods that can construct reliable prediction sets even when dealing with incomplete biological data. As an intermediate step to validate our geometric approach in a controlled environment, we first evaluated the framework on a binary Gram-type classification task before moving on to the full multi-host prediction problem.

2 Background & State of the Art

In this section, we provide the scientific and mathematical background and context for the methods developed in this report. We will begin by introducing bacteriophages and the host-prediction problem in Section 2.1, which motivates the need for computational approaches to find a solution. We will then go over the already existing and state-of-the-art techniques in Section 2.2, progressing from the classical sequence similarity techniques to the much more efficient and modern protein embeddings techniques. We will also look at the limitations that all the currently existing methods have in common, for example the absence of rigorous uncertainty quantification. In the remaining part of the section, 2.6 and 2.5 we will introduce the mathematical framework of conformal prediction and the score-based aggregation method of in [1], on which we will build our work.

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2.1 Bacteriophages and Host-Prediction Problem

Bacteriophages, commonly referred to as phages, are viruses that specifically infect and replicate within bacteria. They are one of the most abundant biological entities on Earth with an estimated population of 10^{31} in the global biosphere [2], and therefore play a very important role in shaping the microbial communities and evolution. In the recent years the interest in bacteriophages has increased due to the growing threat of antimicrobial resistance and the development of phage therapy as an alternate treatment method.

One of the key characteristics of phages is its host range, i.e. the set of bacterial hosts that the phage is capable of infecting. The host range of a phage is not a fixed property, it is determined by molecular interactions between the phage and the bacterial surface structures during the cycle of infection [2], [3]. Each phage attaches itself to the bacterial host, injects its viral genetic material which are able to replicate inside the host. Therefore, some of the phages only infect a few hosts (have a small host range) as they can only infect a limited number of bacterial strains while there are other which are capable of infecting a broad range of bacterias.

Knowing which phage infects which host, can have several applications both in environmental and clinical studies. In clinical, the host range helps us determine how suitable a phage is for designing the treatment for phage therapy (an alternate method to antibiotics). Similarly in environmental studies it is valuable to know the host being infected to understand the ecological interactions between viruses and microbial communities [3]. And it also improves our understanding of viral diversity and evolution.

Host range has its benefits, but at the same time it also brings its own obstacles. In phage therapy, we need to find precisely the phage that infects the given bacterial pathogen which is the cause of the infection. Therefore, we want to be able to identify the correct phage from the thousands of available ones and for that we need a fast reliable host

prediction method. Traditionally, the method of determining the host range used to depend on laboratory experiments in which candidate phages were tested against a number of bacterial hosts; although this method was able to give accurate reliable results but it was extremely expensive and time consuming. Especially with the rapid increase in the phage genomes generated by modern technology, the number of possible phage-host interactions has increased astronomically, and this leads us to the need of having a computational prediction method. We need a way which can complement the lab experiment by providing with a shorter list of candidate phages before we move on to experimental validation [4] or even directly infer the host information directly from a given phage.

We can formulate this host prediction problem as a supervised classification task, given information associated with a phage our objective should be to be able to predict the bacterial host from a given set of candidate hosts, $\mathcal{H} = \{h_1, \dots, h_m\}$. Let ψ represent the phage, and $\mathcal{H}$ be its host label the objective is to construct a rule

$$f : \psi \rightarrow \mathcal{Y}$$

which can accurately predict the unknown host subset $H^* \subseteq \mathcal{H}$ from the phage information available.

2.2 State of the Art

already existing method for the phage host prediction

- Brief overview of existing approaches in machine learning
- Why Bayesian methods are unsuitable here: require distributional assumptions, computationally expensive for large-scale genomic data, posteriors depend heavily on prior choice
- Why calibration methods are insufficient: they produce calibrated probabilities but no formal finite-sample guarantee
- Why conformal prediction is the right choice: distribution-free, finite-sample guarantee, no assumptions beyond exchangeability, computationally lightweight

2.3 State of the Art in Computational Host Prediction

Computational approaches to resolve f have rapidly evolved across three distinct technological paradigms.

2.3.1 Sequence-Similarity and Homology-Based Heuristics

Early computational frameworks operated under the assumption that viral genomes retain significant evolutionary signatures of their hosts. Tools such as **WIsH** [?] utilize Markov

chains to model k -mer frequency trajectories, computing the log-likelihood of a short phage metagenomic contig originating from a host’s genomic background. Concurrently, alignment-driven tools leverage software like **MMseqs2** [?] to cross-reference phage sequences against curated repositories like **INPHARED** [?], seeking localized homology matches. These include prophage sequences integrated into bacterial chromosomes or specific CRISPR spacer matches. While highly precise when a close reference genome exists, these alignment heuristics fail entirely when analyzing the "viral dark matter"—novel phages lacking sequence homology in public databases [4].

2.3.2 Protein-Content and Classical Machine Learning Paradigm

To capture broader functional signals, a second generation of tools shifted focus from raw nucleotide sequences to translated protein profiles. **RaFAH** [?] trains a Random Forest architecture on a large-scale viral protein cluster matrix, assigning host source probabilities based on the collective presence or absence of specific functional open reading frames (ORFs). Similarly, unified pipelines like **iPHoP** integrate host alignment signals, co-abundance profiles, and protein content scores into integrated classifiers. While more robust to sequence divergence than nucleotide alignment, these methods require explicit functional annotations, which are missing for a large percentage of discovered viral genes.

2.3.3 Deep Language Models and Protein Embeddings

The current frontier in host-range prediction leverages self-supervised Large Language Models (LLMs) trained on evolutionary scales of amino acid sequences. Foundation models like **ProtT5** [?] map protein sequences into continuous, dense vector spaces ($\mathbb{R}^d$) that capture complex biochemical, biophysical, and structural features without requiring explicit sequence alignment.

This embedding paradigm has yielded major performance gains:

- **Empathi** [?] utilizes dense protein embeddings within a hierarchical classification setup to perform deep, automated functional annotations of highly divergent phage proteins.
- **SUBLYME** [?], embedded within the PhaLP 2.0 pipeline, applies specialized embedding representations to specifically discover and isolate functional phage lysins from diverse metagenomic data.
- Direct host-range predictors, such as the neural frameworks proposed by Gonzales et al. [?] and **Du et al. (2026)** [?], feed these protein embeddings into deep neural networks to directly map phages to specific host strains, demonstrating that LLM representations can capture the structural constraints governing receptor-binding proteins.

2.4 The Critical Deficiency: Failure of Uncertainty Quantification

Despite their empirical success, all three generations of computational tools share a fundamental, systemic flaw: they operate exclusively as **point predictors**. For an input phage feature vector $\mathbf{x} \in \mathcal{X}$, these algorithms output a soft score vector $\mathbf{s}(\mathbf{x}) = (s_1(\mathbf{x}), \dots, s_K(\mathbf{x})) \in [0, 1]^K$, where $s_k(\mathbf{x})$ is interpreted as a heuristic confidence or empirical probability that the phage infects host k . Crucially, these scores completely lack rigorous **Uncertainty Quantification (UQ)**. They fail to decouple *aleatoric uncertainty* (inherent stochasticity in the biological infection process) from *epistemic uncertainty* (lack of model knowledge due to sparse training data) [1].

In clinical phage therapy, executing choices based on raw point predictions introduces severe risks. If a model yields $s_1(\mathbf{x}) = 0.51$ for Host 1 and $s_2(\mathbf{x}) = 0.49$ for Host 2, a point-prediction rule forces the selection of Host 1, despite the model being highly uncertain. To mitigate this risk, we analyze why standard statistical machine learning approaches fail to provide satisfactory guarantees.

2.4.1 Why Bayesian Formulations Are Unsuitable

A classical mathematical approach to UQ is the Bayesian paradigm, which targets the posterior predictive distribution $P(Y | X, \mathcal{D})$. While theoretically elegant, Bayesian methods are poorly suited for large-scale phage-host modeling for three reasons:

1. **Computational Intractability:** Evaluating the posterior distribution over high-dimensional protein embedding spaces requires complex Markov Chain Monte Carlo (MCMC) sampling or Variational Inference (VI). This is computationally prohibitive when scaled to massive matrices containing millions of bacterial and viral proteins [?].
2. **Prior Sensitivity:** In high-dimensional, sparse biological matrices, the calculated posterior distributions depend heavily on arbitrary choices of prior distributions, which lack biological or statistical justification.
3. **Misspecification Vulnerability:** Bayesian guarantees are strictly asymptotic and rely on the structural assumption that the true data-generating distribution lies within the chosen model class ($\mathcal{M}$ -closed setting). If the underlying model architecture is misspecified, the resulting credible intervals lose their validity.

2.4.2 Why Standard Calibration Methods Are Insufficient

Alternative frequentist methods focus on post-hoc calibration techniques, such as Platt Scaling or Temperature Scaling. These methods solve an optimization problem over an

independent validation set to ensure that the output scores match empirical frequencies, meaning:

$$P(Y = k \mid s_k(X) = \hat{p}) \longrightarrow \hat{p} \quad \text{as } n \rightarrow \infty \quad (2.4.1)$$

While calibration ensures that a confidence score of 0.90 aligns with a 90% long-run empirical success rate on average, it suffers from severe limitations:

1. **No Finite-Sample Guarantees:** Calibration metrics (such as the Expected Calibration Error, ECE) only provide asymptotic regularities. They offer no statistical safeguards when working with small, finite datasets or highly specialized clinical isolates.
2. **Retention of the Point Prediction Paradigm:** Calibration alters the scale of the scalar outputs but does not change the core prediction format. It does not provide a mathematical mechanism to construct a discrete predictive set $C(\mathbf{x}) \subseteq \mathcal{Y}$ that controls the exact instance-level error rate.

This methodological gap demonstrates the need for a framework that is computationally lightweight, completely distribution-free, and capable of rendering exact, finite-sample statistical guarantees. This directly motivates our development of a **conformal prediction** pipeline.

2.5 Conformal Prediction

2.5.1 Setup and nonconformity scores

- define exchangeability, define $s(\mathbf{x}, y)$ -> explain the intuition

2.5.2 prediction set and coverage guarantee

- define $C(\mathbf{x})$

2.5.3 Split conformal prediction

- why we use it, the train/calibration/test separation, validity still holds

2.6 The CSA Method

2.6.1 Motivation

- from scalar to multivariate nonconformity scores -> why a scalar threshold is insufficient when the predictor produces a vector

2.6.2 Two-stage calibration

S_1 for shape discovery, S_2 for size scaling -> what each stage does and why they must be disjoint

2.6.3 The envelope concept

- a region $E \subset \mathbb{R}^K$ learned from S_1 , scaled by $\hat{t}$ calibrated on S_2

2.6.4 Limitations of the original method

- iterative search for $\hat{t}$, convexity assumptions, no class-conditional guarantee -> each of these is addressed in Chapter 3

3 Methods

3.1 Problem Formalization

- Input: a phage represented by a score vector in $[0, 1]^K$ where K is the number of functional dimensions
- Define the label space $\mathcal{Y}$ (finite set of classes)
- Goal: construct a prediction set $C(x) \subseteq Y$ with the conformal guarantee
- The one-envelope design: build one single envelope on all training phages under their true-label NC transformation, rather than separate envelopes per class

3.2 Non Convex envelope for classification

- In the regression setting of the original CSA paper, envelopes are typically convex because the score space is unbounded and the data does not cluster into separated groups
- In our classification setting, the score space is bounded in $[0, 1]^K$ and — crucially — true NC vectors for different classes occupy distinct, well-separated regions of this space (example)
- In the classification, produce tighter prediction sets without sacrificing the coverage guarantee
- The guarantee still holds because it depends only on the exchangeability of the data and the quantile construction, not on the shape of the envelope

3.3 Score Transformation

- General formulation: for each class $y \in \mathcal{Y}$, define a transformation $\phi_y: [0, 1]^K \rightarrow [0, 1]^K$ that maps the raw score vector to an conformal score vector such that a true infector of class y is near the origin
- Property we require: $\phi_{i_y}(\mathbf{s})$ is small when $\mathbf{s}$ is consistent with label y , large otherwise
- we make a single unified envelope coherent across all classes

3.4 Envelope Constriction

3.4.1 CSA Two-Stage Calibration

- Training vectors are split into two disjoint halves S_1 and S_2 (50/50 random split).
- S_1 (shape discovery): used to learn the geometry of the envelope.
- S_2 (size scaling): used to calibrate the scaling factor $\hat{t}$ that determines the final size of the envelope.

- The two stages must use disjoint data to preserve the exchangeability argument underlying the coverage guarantee.

3.4.2 τ Computation

- Define $\tau(\mathbf{s}) = \inf\{t > 0 : \mathbf{s} \in t \cdot E\}$, the minimum scaling factor needed to include point $\mathbf{s}$ in the scaled envelope E .
- The global scaling factor is then determined by the $(1 - \alpha)$ quantile of the τ values over S_2 :

$$\hat{t} = \tau_{(k)} \quad \text{where} \quad k = \lceil (|S_2| + 1)(1 - \alpha) \rceil$$

where $\tau_{(k)}$ denotes the k -th smallest value of τ in S_2 .

- Key contribution: for each of the three envelope geometries, $\tau(\mathbf{s})$ has a closed-form expression computable in a single pass — no iterative search required.
- Computational benefit: the entire calibration step reduces to one forward pass over S_2 followed by a sort.
- Contrast with the original CSA paper which uses iterative binary search.

3.4.3 Radial Method

3.4.4 Strip Method

3.4.5 1D

3.4.6 Class-Specific $\hat{t}$ Calculation

3.4.1 — CSA Two-Stage Calibration

3.5 $\hat{t}$ Calculation

- class conditional (why and how)

4 Data

4.1 Data Description

- The two files: protein-level score matrix (1,853,074 proteins $\times$ 40 functional score columns, values in $[0,1]$, NaN when protein not annotated with that function) and meta-data file (same 1,853,074 rows, columns: accession, host, host_type, split)
- Scale of the dataset: 18,474 unique phages after aggregation
- The split column: integer 0–4 encoding a genome-cluster-based partition — phages in the same genome cluster are always in the same fold, ensuring test phages are genuinely different from training phages at the sequence level
- Class distribution: gram-neg vs gram-pos vs unknown
- The NaN structure: NaN is structural missingness, not noise — a protein that was not annotated with a given function simply has no score for that function, which is different from a score of zero

4.2 Protein to Phage Aggregation

- The raw data is at the protein level but conformal prediction operates at the phage level — aggregation is needed
- NaN-aware mean pooling: for each phage and each functional score column, compute the mean across all proteins of that phage that have a non-NaN value for that column. Proteins with NaN for a given column are excluded from the mean for that column only
- Why not replace NaN with zero before pooling: treating an unannotated protein as having zero evidence would push the phage’s aggregated score artificially toward zero, misrepresenting its biology
- After aggregation: result is a matrix of shape (number of phages) $\times$ 40, still with NaNs in cells where none of the phage’s proteins were annotated with that function

4.2.1 NaN handling after aggregation

- After mean pooling, some phages still have NaN in certain columns -> this happens when none of their proteins were annotated with a particular function
- Strategy: fill remaining NaNs using column-wise medians computed strictly on the training set (never the test set, to prevent data leakage) - The median cap: any training median above 0.6 is capped at 0.5 -> this prevents an imputed value from falsely pushing a phage toward a particular class - any remaining NaNs after median imputation are filled with 0.5 (neutral value)

4.3 Train-Test Split

- Standard split: split 0 is the held-out test set, splits 1–4 form the training set
- The contamination problem: explain the protein cluster (pc) column -> a protein's cluster may include proteins from phages in different folds. The models that produced the scores were trained on protein clusters, so a protein from a training-fold phage may share a cluster with a test-fold protein, introducing subtle leakage
- The masking fix: the file `all_random_pc_splits.pkl` encodes for each protein and each (host, function) model which fold that protein was assigned to. Setting scores to NaN where the protein was not in the test set of the relevant model, before aggregation, removes the contaminated predictions. The existing NaN-aware mean pooling then handles these naturally -> this is the "missing value" framing

5 Experiments and Results

5.1 Synthetic Data Validation

5.1.1 Purpose and setup

- Why synthetic validation before real data: allows controlled verification that the coverage guarantee holds exactly, with known ground truth distributions
- General setup: simulate score vectors from known distributions, apply all three envelope methods, verify empirical coverage matches theoretical target $1 - \alpha$

5.1.2 2D Experiments

- Three scenarios tested in 2D ($K=2$):
 - Uncorrelated scores: two independent uniform or Gaussian dimensions
 - Correlated scores: scores with positive correlation -> tests whether the envelope adapts to the correlation structure
 - Asymmetric predictors: one informative dimension and one uninformative dimension -> tests whether the envelope correctly ignores the noisy dimension
- For each scenario: show the envelope shape visually, report empirical coverage across multiple runs, confirm it matches $1 - \alpha$
- Key observation from 2D: the radial and strip envelopes are visibly non-convex and wrap tightly around the data, while collapsed 1D produces a simple interval

5.1.3 Extension to K Dimensions

- Repeat the experiments for $K \in 2, 5, 10, 20$ with a single host
- Show that all three methods maintain coverage $\geq 1 - \alpha$ across all dimensionalities
- Discuss the effect of increasing K on envelope quality: radial method becomes sparser (fewer S1 points per sector), strip method has more dimension pairs to condition on, collapsed 1D is unaffected
- Show that the analytical τ computation gives identical results to what iterative search would give, confirming correctness

5.1.4 Multiple Hosts

- Extend to a simulated multi-class setting with multiple hosts ($|\mathcal{Y}| = 6, K = 5$)
- Each host's true score distribution is a distinct cluster in $[0, 1]^K$
- Verify that the single unified envelope correctly assigns the true host to the prediction set with probability $\geq 1 - \alpha$
- Show 3D visualisation of the envelope for $K=3$

5.2 Gram Type Classification

5.2.1 Experimental Setup

- Data: split-0 phages only (80% calibration, 20% test), unknown host types excluded
- Parameters: $\alpha = 0.1$, $M = 200$ directions (radial), $M=20$ bins (strip), 10 independent runs to estimate variance
- The NC transformation used: keep scores as-is for one class, flip with $1 - s$ for the other

5.2.2 Results

- table: overall coverage, per-class coverage, average set size, empty set rate, size-2 rate, singleton accuracy std across 10 runs for all three methods
- Bar charts: set size distribution, overall and per-class coverage

5.3 Host Classification

6 Conclusion and Future Work

6.1 Summary.....

6.2 Limitations

- gram-pos coverage -

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